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State sensing of bubble jet flow based on acoustic recognition and deep learning

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ABSTRACT

This study covers the accidental generation of bubble jet flow caused by steam generator (SG) tubes damaging in sodium-cooled fast reactors (SFRs). The main objective of this study is to develop a novel state sensing method of bubble jet flow based on acoustic recognition and deep learning. Prior to the application of this method to actual SFRs, we utilize air and water as simulant fluids in order to perform the proof of concept. This study is divided into three phases. The first phase is the acquisition and analysis of pipe flow sound and bubble jet flow sound, each of which simulates the normal and anomaly sound from SG tubes in SFRs. The second phase is the preprocessing of acoustic signals and feature extraction. The third phase is the building of deep learning models and performance evaluation. As a result, every of our proposed models could distinguish between pipe flow sound and bubble jet sound with an accuracy of almost 100.00%, and the best model could classify pipe flow sound and three types of bubble jet flow sound with an accuracy of 99.76%. This result suggests that the acoustic recognition with deep learning has great potential to sense the state of bubble jet flow in actual SFRs.

1. Introduction

Nuclear power generation attracts the world's attention once again since the Paris Agreement was adopted in 2015 and many countries began to aim for the realization of carbon neutrality. Although thermal power generation, one of the main power generation methods, emits a large amount of greenhouse effect gases such as carbon dioxide, nuclear power generation emits them only under construction (Sadekin et al., 2019). Considering this property, the EU recognizes nuclear energy as clean, and electricity generated by nuclear energy accounts for about 42% of low-carbon electricity in the EU now. Thus, nuclear power has the potential to be a solution for achieving carbon neutrality, and there are movements to accelerate innovation in nuclear technology such as the Advanced Reactor Demonstration Program (ARDP) supported by the United States Department of Energy (DOE).

One of the next-generation nuclear reactors is the SFR, which can breed nuclear fuels while generating electricity because of its characteristics of a fast-spectrum reactor and closed fuel recycle system (Aoto et al., 2014). In SFRs, if a heat transfer tube is damaged and a crack occurs, water in the heat transfer tube leaks to liquid sodium in the secondary cooling system. This causes the sodium–water reaction, which widens the leakage hole and turns the phases of the water leakage: small, intermediate, and large (Judd et al., 1990; Matsuura et al., 2007). With the progression of three phases, the sodium–water reaction causes effects such as self-wastage, single-target wastage, and multi-target wastage (Hori, 1980). In particular, the last two effects are caused by the two-phase critical flow with high temperature and strongly alkaline, and this jet flow damages surrounding heat transfer tubes. Accidents with the sodium–water reaction in SFRs were reported in past, and dozens of tubes were damaged in some cases (Judd et al., 1990; Hartwell, 1963; Poplavsky, 1990; Alanche et al., 1990). The root causes of them were penetrating cracks generated by fluid vibration, weld defects, and heat exhaustion. However, serious accidents in the SG can be prevented by early-stage leak detection before the progression of phases. To develop early-stage leak detection systems, some studies focus on acoustic methods with the consideration of their advantage of a short time response (Greene et al., 1984; Brunet et al., 1988; Chikazawa, 2010).

In the 21st century, emerging technologies are developed with the breakthrough of computers such as Artificial Intelligence (AI) (Ghahramani, 2015), Internet of Things (IoT) (Atzori et al., 2010), and big data (McAfee and Brynjolfsson, 2012). Among them, deep learning, one of the AI techniques, is developing at an extraordinary speed. It is based on neural networks, which mimic the mechanism of human nerve cells named neurons. Deep learning models consist of multilayer neural

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networks. They are trained by using the backpropagation algorithm and changing their internal parameters. Through repetitive training, they improve the accuracy or reduce the loss between actual and predicted values, i.e., they get smarter over time just like the human brains. Deep learning models trained in this way sometimes exceed human cognitive ability. This technique dramatically improves the state of art in various recognition fields including sound recognition (LeCun et al., 2015). Some studies utilize fully connected neural networks (FCNNs), basic deep learning models, and input one-dimensional (1D) features such as amplitude or power spectrums generated by fast Fourier transform (FFT) (Ueki and Ara, 2021; Bae et al., 2018). In addition, more and more studies focus on convolutional neural networks (CNNs), best ways of image recognition, and input two-dimensional (2D) features such as the graphs of spectrums as images (Bae et al., 2018), spectrograms generated by FFT (Raza et al., 2019; Satt et al., 2017), or scalograms generated by wavelet transform (Yaseen et al., 2018; Loey and Mirjalili, 2021). Also, deep learning is developing in the identification of the states of two-phase flow. For example, Poletaev et al. (2020) captured the images of bubbles in the two-phase jet flow and showed that the methods using neural networks could correctly judge the presence of bubbles and determine the geometric center of bubbles. However, this method based on the optical image recognition of bubbles is not applicable in SFRs because the inside of SGs cannot be visualized. To this problem, we focus on acoustic methods with the consideration of properties of sound propagation: sound also propagates from its source through structures such as piping systems. To the identification of bubble flow regimes, Zhang et al. (2021) developed a method based on deep learning and visual representation images of acoustics. The result showed that the proposed method reached an accuracy of 99.67% for the identification of three types of bubble flow: homogeneous, transition, and heterogeneous flow. However, little is known about the state sensing of bubble jet flow assumed in accidents in actual SFRs such as those emitted at the speed of sound, or with slightly different emission hole diameters by a method based on acoustic recognition and deep learning.

Based on the above, this study covers the accidental generation of bubble jet flow in the accident of SG tubes damaging in SFRs, and the aim of this study is to develop a novel method to sense the states of bubble jet flow based on acoustic recognition and deep learning. Our proposed method can be applied to invisible circumstances including SGs. Also, this method has the possibility to detect anomalies in the early stage by the acoustic method with a short response time and analyze the progression of the accident by deep learning, which can lead to the safe operation of SFRs. The paper begins with the description of experiments on the acquisition of pipe flow sound and bubble jet flow sound, each of which simulates the normal and anomaly sound from SG tubes in SFR. This is followed by the presentation of proposed feature extraction methods and generated features. Finally, proposed deep learning models and performance evaluation are shown.

2. Experiments of sound acquisition

2.1. Experimental arrangement

Pipe flow sound simulates normal sound from the SG. Fig. 1 shows a experimental arrangement for sound acquisition of pipe flow. A pipe made of SUS304 is supported horizontally, and filled with water. Its outer and inner diameters are d_1 = 165.2 mm and d_2 = 158.2 mm, respectively. A hydrophone (ACO TYPE7800, Japan) is located 200 mm below the pipe in a water tank, and about $15d_2$ downstream from the pipe inlet. The dimension of the water tank is 300 (W) × 450 (D) × 600 (H). The hydrophone records sound at a sampling rate of 44.1 kHz for 1 min. To conduct the experiment under conditions close to the Reynolds number of actual SFR ($Re = 2.7 \times 10^5$), the flow rate is set at 125 m^3/h ($Re = 3.2 \times 10^5$).

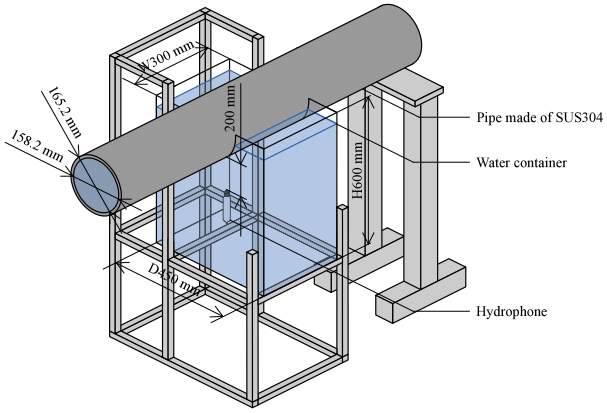


Fig. 1. Experimental arrangement for sound acquisition of pipe flow.

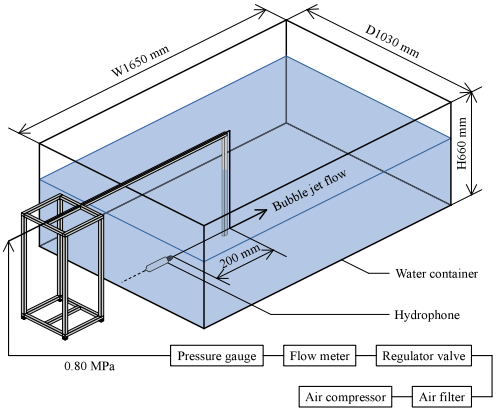


Fig. 2. Experimental arrangement for sound acquisition of bubble jet flow.

Table 1
Experimental condition for sound acquisition of bubble jet flow sound.

No.	Nozzle diameter mm	Pressure MPa	Air temperature K	Mass flow rate kg/s
1	0.15	0.80	294.2	1.53×10^{-5}
2	0.17	0.80	294.5	2.65×10^{-5}
3	0.20	0.80	294.3	4.13×10^{-5}

Bubble jet flow sound simulates anomaly sound from the SG. Fig. 2 shows a experimental arrangement for sound acquisition of bubble jet flow and parameters in each condition are shown in Table 1. Air is compressed to 0.80 MPa (absolute pressure) by an air compressor (AIRREX FX8701, Japan), and released from a nozzle into water through a filter (KEYENCE FD-C5, Japan), a regulator valve (KEYENCE FD-C1, Japan), a flow meter (KEYENCE FD-A10, Japan), and a pressure gauge (Naganokeiki GC31, Japan). These components are connected by 1/4 inch tubes. The dimension of the water tank is 1650 (W) × 1030 (D) × 660 (H). Three types of nozzles are prepared, each with the nominal diameter of 0.15 mm, 0.17 mm, and 0.20 mm. Nozzle holes are drilled perpendicular to the direction of air flow, and they are not perfect circles because of micromachining, and actual diameters measured by an optical microscope are from 0.16136 to 0.16538 mm for the nominal diameter of 0.15 mm, from 0.18002 to 0.18336 mm for the nominal diameter of 0.17 mm, and from 0.22826 to 0.23415 mm for the nominal diameter of 0.20 mm. The hydrophone same as shown previously is located 200 mm behind bubble jet flow, and records bubble jet flow sound from each nozzle at a sampling rate of 44.1 kHz for 1 min. Also, the states of bubble jet flow are captured by an action camera (Apexcam X60-pro, China). Additionally, the nozzle pressure

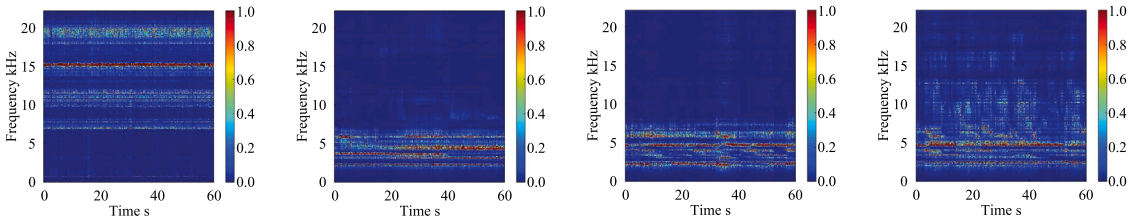


Fig. 3. Spectrograms of measured sound signals (from left to right: pipe flow sound, bubble jet flow sound from nozzles with diameters of 0.15, 0.17, and 0.20 mm).

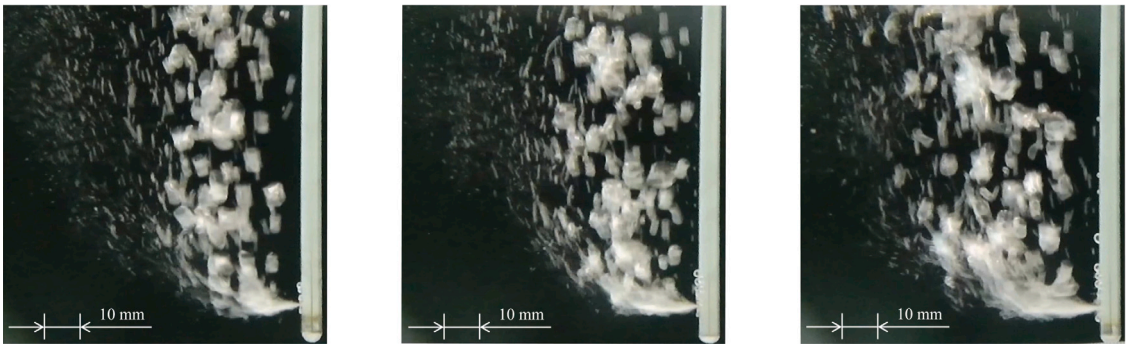


Fig. 4. Snapshots of bubble jet flow (from left to right: bubble jet flow from nozzles with diameters of 0.15, 0.17, and 0.20 mm).

ratio is expressed by the following equation (Pathan et al., 2019):

$$\frac{p_0}{p} = \left(1 + \frac{1}{2}(\gamma - 1)Ma^2\right)^{\gamma/(\gamma-1)} \tag{1}$$

where γ is the specific heat ratio, Ma is the Mach number, p and p_0 denote the pressure at the nozzle outlet and the stagnation point, respectively. Considering $\gamma = 1.4$ for air and $Ma = 1$, Eq. (1) becomes the following equation:

$$p_0 = 1.893p^* \tag{2}$$

where p^* denotes the pressure at the sound velocity point. p_0 is more than 1.893 times p^* in our experiment, so the air flow is choked at the nozzle outlet, which is consistent with the conditions expected in the actual accident.

2.2. Experimental result

Measured pipe flow sound and three types of bubble jet flow sound are firstly processed by a high-pass filter with a stopband of 900 Hz or less and a passband of 1000 Hz or more to remove noise in the low frequency band. Through the fast Fourier transform, spectrograms of measured sound signals are obtained as shown in Fig. 3. In these spectrograms, the signal powers are weighted to show the spectrum distributions clearly. From these results, the pipe flow sound has several peak frequencies including about 7 kHz, 10 kHz, 14 kHz, and 18 kHz. We speculate that peak frequencies of the pipe flow sound are based on the coupled waves of pipe walls and internal water. Also, the three types of bubble jet flow sound have a characteristic frequency band of 1500 Hz to 5000 Hz. We speculate that the bubble jet flow sound is based on the vibration of bubbles. The model of a single bubble vibration in the radial direction is proposed by M. S. Plesset and A. Prosperetti, and the natural frequency of the bubble vibration is described as follows (Plesset and Prosperetti, 1977):

$$f = \frac{1}{2\pi}\sqrt{3\kappa\frac{p_{g0}}{\rho r^2} - \frac{2\sigma}{\rho r^3}} \tag{3}$$

where r denotes the bubble radius, ρ and σ are the density and surface tension of liquid respectively, κ is the specific heat ratio of gas, and p_{g0} is the gas pressure at the surface of the bubble in a state of rest.

Also, the snapshots of three types of bubble jet flow are shown in Fig. 4. As shown in these snapshots, the reach of bubble jet flow becomes longer with an increase of the nozzle diameter. In the area close to the nozzle, there are large bubbles with the centimeter scale. On the other hand, there are small bubbles with the millimeter scale in the area distance from the nozzle. Considering a characteristic frequency band of 1500 Hz to 5000 Hz for the bubble jet flow sound, we speculate that the vibrations of bubbles with a radius from 0.6 mm to 2.2 mm are the main components of the bubble jet flow sound based on Eq. (3). The bubbles with a radius in this range are observed in the experiment as shown in Fig. 4.

3. Feature extraction

3.1. Proposed feature extraction methods

Acoustic features prepared for deep learning are two types: root mean square (RMS) and frequency-pattern features. Frequency-pattern features consist of amplitude spectrum (AS), power spectrum (PS), and spectrogram features. RMS, amplitude spectrum, and power spectrum features each have a 1D feature, and two types of 2D features: line mapped (LM) and domain mapped (DM) features. The LM features are generated with reference to the previous study (Bae et al., 2018), and the DM features are what we newly proposed. Finally, we prepare a total of ten features: (1) 1D RMS, (2) 2D RMS-LM, (3) 2D RMS-DM, (4) 1D AS, (5) 2D AS-LM, (6) 2D AS-DM, (7) 1D PS, (8) 2D PS-LM, (9) 2D PS-DM, (10) 2D Spectrogram. The following shows each method of feature extraction.

3.1.1. Generation of RMS-pattern features

Fig. 5 shows the process of generating RMS-pattern features. First, an acoustic signal x is preprocessed by the high-pass filter (HPF) same as shown in the previous section for noise reduction. For path A, $y_{\mathrm{rmsA}} = [y_1, y_2, \ldots, y_T]$ is created by direct calculation of the RMS level according to the following equation:

$$y_n = \left\{\sum_{k=1}^{N_w} {x_{n+k}}^2\right\}^{1/2} \tag{4}$$

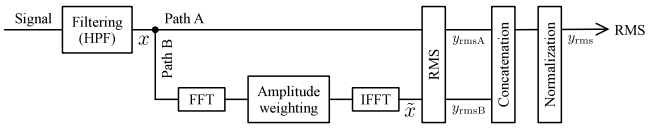


Fig. 5. Process of generating RMS-pattern features.

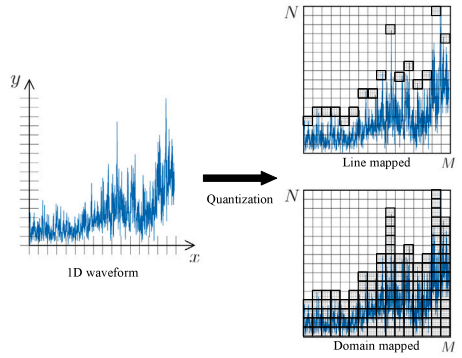


Fig. 6. 2D imaging of 1D waveforms.

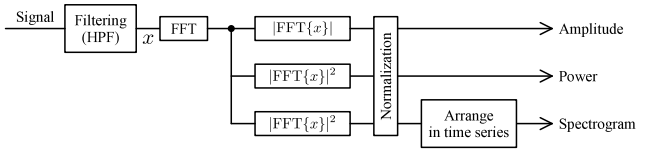


Fig. 7. Process of generating frequency-pattern features.

where T determines the length of y_{rmsA}, and N_w is the order of a moving average. For path B, an amplitude-adjusted $\tilde{x}$ is calculated by the following equation:

$$\tilde{x} = \mathrm{IFFT}\{W(f) \cdot \mathrm{FFT}\{x\}\} \tag{5}$$

where FFT and IFFT denote fast Fourier transform and inverse fast Fourier transform, respectively. $W(f)$ is defined as the following equation:

$$W(f) = \begin{cases} w & (f_L \le f \le f_H) \\ 0 & (f < f_L,\ f_H < f) \end{cases} \tag{6}$$

where f_L and f_H are values which determine the range of windowing. y_{rmsB} is created by replacing x with $\tilde{x}$ in Eq. (4) and calculating it. The normalized 1D RMS (y_{rms}) is generated by concatenation of two RMS-level feature vectors as follows:

$$y_{\mathrm{rms}} = \frac{1}{y_{\max}}[y_{\mathrm{rmsA}},\ y_{\mathrm{rmsB}}] \tag{7}$$

where $y_{\max}$ is the maximum of y_{rmsA} and y_{rmsB}.

In the next step, 2D RMS-LM and 2D RMS-DM are generated as shown in Fig. 6. First, the original graph of 1D waveform is discretized by a 2D $M \times N$ grid, such as M-levels along the x-axis and N-levels along the y-axis. The values of y_{rms} are quantized with the same N-levels with respect to the corresponding M-intervals. The LM features are generated by mapping only the quantized values. In addition, the domain-like mapping generates light and dark areas, which may enable classifiers to recognize clearly the change of RMS values for the RMS-pattern features and the peak for the frequency-pattern features. Based on this idea, we newly propose the DM features, which are generated by mapping the quantized values or less as shown in Fig. 6.

3.1.2. Generation of frequency-pattern features

Fig. 7 shows the process of generating frequency-pattern features. First, a signal x in time-series is preprocessed by the HPF, and converted to a signal in frequency-series by FFT. 1D AS and 1D PS are generated by calculating the absolute values of FFT$\{x\}$ and the square of the absolute values of FFT$\{x\}$, respectively, and normalization. 2D AS-LM and DM, and 2D PS-LM and DM are generated by the same method shown in the previous section. Additionally, arranging of normalized PS in time series generates the 2D Spectrogram, in which the horizontal axis represents time, the vertical axis represents frequency, and the intensity of sound is represented by color.

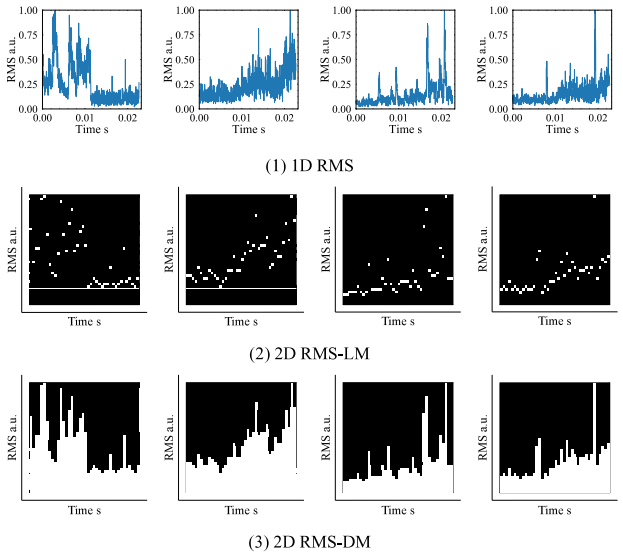


Fig. 8. Examples of RMS-pattern features (from left to right: pipe flow sound, bubble jet flow sound from nozzles with diameters of 0.15, 0.17, and 0.20 mm).

3.2. Generated features

Examples of RMS-pattern features are shown in Fig. 8. The length of 1D RMS is 1000. The shape of 2D RMS-LM and DM is $40 \times 40 \times 1$, which means that the height and width of images are 40 pixels with gray-scale. Considering the characteristic frequency band of bubble jet flow sound, the thresholds of the window function in Eq. (6) are set to $f_L = 1500$ Hz and $f_H = 5000$ Hz. Also, we fix the weight value to $w = 5$. Therefore, at the middle of time-axis, the values of RMS are attenuated for pipe flow sound, and amplified for bubble jet flow sound.

Fig. 9 shows examples of frequency-pattern features. The lengths of 1D AS and PS are 1000. The shapes of 2D AS-LM and DM, and 2D PS-LM and DM are $40 \times 40 \times 1$, which means that the height and width of images are 40 pixels with gray-scale. Additionally, the shape of 2D Spectrogram is $40 \times 40 \times 3$, which shows that the height and width of images are 40 pixels with RGB-scale.

Table 2 shows the prepared data through the feature extraction mentioned above. The numbers in the Table 2, from left to right, represent the number of training data, validation data, and test data divided for deep learning. All extracted data were split into training, validation, and test dataset with portions of 80%, 10%, 10%, respectively.

4. Deep learning

4.1. Proposed deep learning models

Deep learning models employed in the present study are FCNN and CNN models as shown in Fig. 10. The FCNN model handles 1D features and consists of an input layer, five affine layers, and an output layer. The input shape is 1000×1, and the numbers of nodes in hidden layers are 1600, 800, 400, 200, and 100 in the forward direction. The

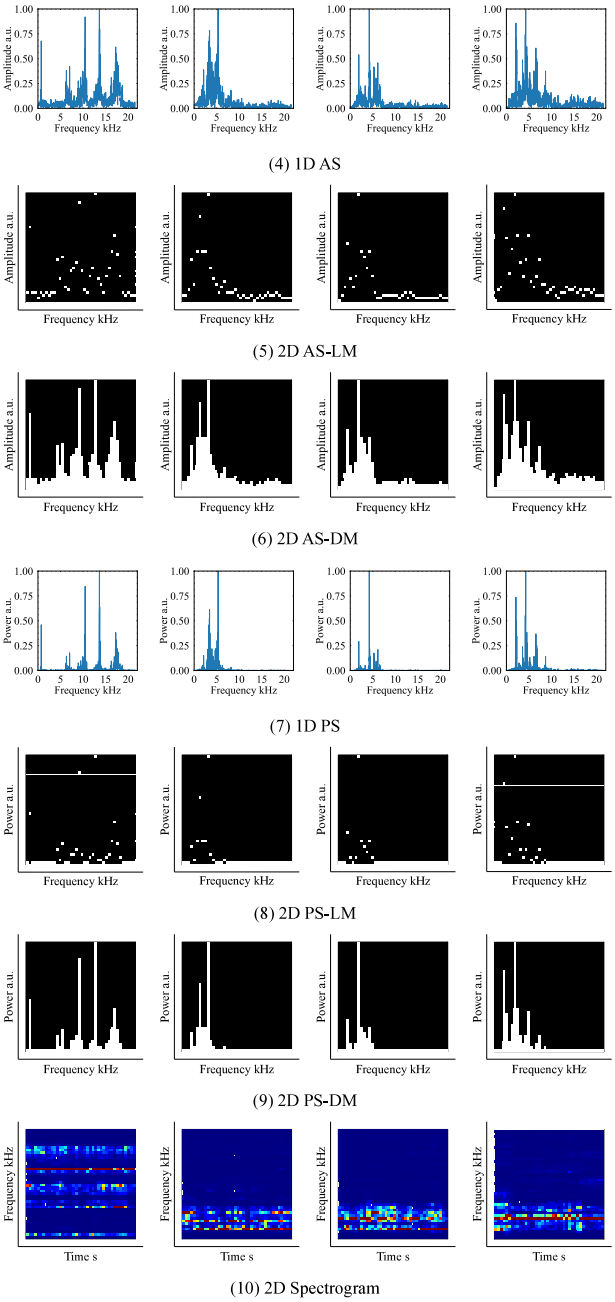


Fig. 9. Examples of frequency-pattern features (from left to right: pipe flow sound, bubble jet flow sound from nozzles with diameters of 0.15, 0.17, and 0.20 mm).

activation function for the hidden layers is the rectified linear unit (ReLU) function:

$$f(x) = \begin{cases} x & (x \geq 0) \\ 0 & (x < 0) \end{cases} \tag{8}$$

The ReLU function performs best in most cases of deep learning (Sharma et al., 2020). Also, the activation function for the output layer is the softmax function:

$$f(x_i) = \frac{\exp(x_i)}{\sum_{k=1}^{n} \exp(x_k)} \tag{9}$$

where n is the number of nodes in the output layer, and x_k $(k = 1, \ldots, n)$ denotes the input vector. The softmax function returns the probability

Table 2
Prepared datasets of each feature (train, validation, test).

Feature		Pipe flow sound	Bubble jet flow sound		
			0.15 mm	0.17 mm	0.20 mm
(1)	1D RMS				
(2)	2D RMS-LM		(960, 120, 120)		
(3)	2D RMS-DM				
(4)	1D AS				
(5)	2D AS-LM				
(6)	2D AS-DM		(1920, 240, 240)		
(7)	1D PS				
(8)	2D PS-LM				
(9)	2D PS-DM				
(10)	2D Spectrogram		(96, 12, 12)		

of each class (Sharma et al., 2020). In addition, the calculation between the previous layer with i nodes and the next layer with j nodes is as follows:

$$(x_{\text{next, in}})_j = w_{ji}(x_{\text{prev, out}})_i \tag{10}$$

where $(x_{\text{next, in}})_j$ and $(x_{\text{prev, out}})_i$ are the input vector of the next layer and the output vector of the previous layer, respectively. w_{ji} is the weight matrix between the previous layer and next layer. Through updating the values of w_{ji} by the backpropagation algorithm, FCNN models improve their accuracy and loss.

The CNN architecture handles 2D features and consists of an input layer, three convolution layers, a max-pooling layer, an affine layer, and an output layer. The input shapes are $40 \times 40 \times 1$ for images with gray-scale, and $40 \times 40 \times 3$ for images with RGB-scale. The kernel shape of all convolution layers is 3×3, and the numbers of filters are 16, 32, and 64 in the forward direction. The convolution of the input image and kernel is calculated as shown in the equation:

$$(x_{\text{out}})_{i,j} = \sum_n \sum_m k_{n,m}(x_{\text{input}})_{i+n,j+m} \tag{11}$$

where x_{out} and x_{in} are the output and input values of the convolution. $k_{n,m}$ denotes the kernel, which extracts the feature of images. Also, in the max-pooling layer, the maximum values in each 2×2 kernel are extracted. The kernel and weight matrix are updated in training sessions. Similar to the FCNN model, the activation function is the ReLU function for the hidden layers and the softmax function for the output layer. Because the shapes of the output layers are set to be 4×1, both of FCNN and CNN models are constructed to classify the four types of sound: pipe flow sound, and bubble jet flow sound from nozzles with diameters of 0.15, 0.17, and 0.20 mm.

4.2. Performance evaluation and discussion

The FCNN and CNN models are constructed, trained, validated, and tested with the TensorFlow framework. The hyperparameters such as the batch size, the number of epochs, and the learning rate are shown in Table 3. The learning algorithm is the Adam optimizer (Kingma and Ba, 2015), and the loss function is the cross-entropy loss function:

$$L = -\sum_{k=1}^{n} (x_{\text{true}})_k \log(x_{\text{pred}})_k \tag{12}$$

where n denotes the number of classes. Also, $(x_{\text{true}})_k$ and $(x_{\text{pred}})_k$ $(k = 1, \ldots, n)$ are true values and predicted values, respectively. In this study, although the number of epochs is set to be 50, the best model with the highest accuracy or the lowest loss is saved in the 50 training sessions, then saved model is evaluated using the test data. Moreover, we perform the K-fold cross-validation (Refaeilzadeh et al., 2016) to evaluate the generalization performance of each model. We set the value of K to be 10, therefore, the total data is divided into 10 blocks, and each block is used for test data once.

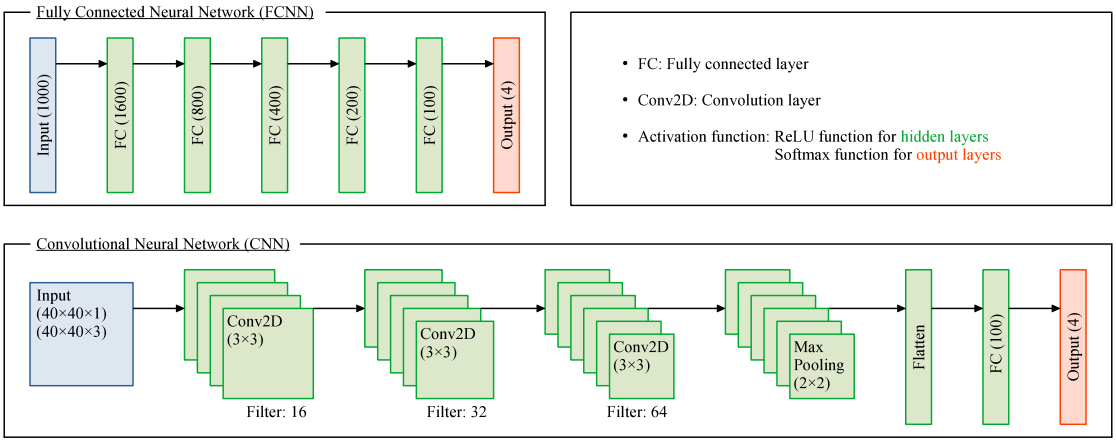


Fig. 10. Architectures of proposed deep learning models.

Table 3
Setup of deep learning.

Model	Batch size	Epoch	Learning rate	Optimization algorithm	Loss function
FCNN CNN	20	50	0.0001	Adam	Cross entropy loss function

The classification results of deep learning models with each feature are shown as confusion matrixes in Fig. 11. In the confusion matrixes, horizontal and vertical directions represent the true and predicted classes, respectively. Deep learning models with features except (2) 2D RMS-LM, (3) 2D RMS-DM, (5) 2D AS-LM, and (6) 2D AS-DM can distinguish between pipe flow sound and bubble jet flow sound with an accuracy of 100.00%. From this result, deep learning models with 1D features and spectrograms are suitable for the distinction between pipe flow sound and bubble jet flow sound. This is because the amount of values in the frequency axis for 1D features is larger than that of 2D image features with gray-scale (one thousand values for 1D features and forty values for 2D image features with gray-scale). Another reason is that the spectrogram has three important components: time, frequency, and signal power, which can perform the 3D display. Although the size of 2D images is set to 40 × 40 in this study, the larger images may improve the performance.

Fig. 12 shows the average accuracy of deep learning models. Deep learning models with frequency-pattern features show higher accuracy than those with RMS-pattern features. This is because RMS-pattern features have RMS values in the time domain, on the other hand, frequency-pattern features have values in the frequency domain, which does not change over time. Also, the best model is built with (4) 1D AS, which reaches an accuracy of 99.76%. This result shows that the FCNN model with (4) 1D AS has not only the ability to detect the bubble jet flow, but also sense its state with high accuracy. Moreover, comparing the LM and DM features, every model with the DM features shows higher accuracy than the LM features, which demonstrates that the performance of models is improved by using the DM features we newly proposed. We speculate that this result is based on the difference in the process of convolution layers. To compare operations in using the LM features and DM features, we consider the convolution of an image with a shape of 5 × 5 and a kernel as a simple example. For the LM feature, the convolution is calculated as follows:

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} k_{1,1} & k_{1,2} & k_{1,3} \\ k_{2,1} & k_{2,2} & k_{2,3} \\ k_{3,1} & k_{3,2} & k_{3,3} \end{bmatrix} = \begin{bmatrix} k_{2,2} & k_{2,1} & 0 \\ k_{1,2} & k_{1,1} & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{13}$$

For the DM feature corresponding to this LM feature, the convolution is described as follows:

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} k_{1,1} & k_{1,2} & k_{1,3} \\ k_{2,1} & k_{2,2} & k_{2,3} \\ k_{3,1} & k_{3,2} & k_{3,3} \end{bmatrix} = \begin{bmatrix} k_{2,2}+k_{3,2} & k_{2,1}+k_{3,1} & 0 \\ k_{1,2}+k_{2,2}+k_{3,2} & k_{1,1}+k_{2,1}+k_{3,1} & 0 \\ k_{1,2}+k_{2,2}+k_{3,2} & k_{1,1}+k_{2,1}+k_{3,1} & 0 \end{bmatrix} \tag{14}$$

where 0 and 1 denote the black and white regions respectively, and $k_{i,j}$ $(i, j = 1, 2, 3)$ represents the kernel. As shown in Eqs. (13) and (14), the amount of kernel values in the convolution for the DM features is larger than that for the LM features because of the difference in mapping methods. For this reason, deep learning models with the DM features can capture more features such as the changes of RMS values and the peaks of spectrums than those with the LM features.

We have developed a novel state sensing method of bubble jet flow based on acoustic recognition and deep learning, which has the ability to detect anomalies in the early stage and to analyze the progression of the accident. However, this method with passive acoustic sensing can be unavailable depending on the locations of sensors in circumstances where bubbles are dominant and sound propagation is impeded. Although the present study does not focus on these circumstances but on the early stage of the water leakage accident in SFRs, we also have to introduce the active acoustic sensing method, which evaluates the attenuation of sound pressure by passing through bubbles.

5. Conclusion

The present study has covered the accidental generation of bubble jet flow from SG tubes in the SFR, and we have developed a novel method of state sensing of bubble jet flow based on acoustic recognition and deep learning by using air and water as simulant fluids to perform the proof of concept. In the phase of experiments, we have acquired pipe flow sound and three types of bubble jet flow sound. In the phase of deep learning, the FCNN and CNN models and ten types of features are prepared, and we have conducted the performance evaluation based on the accuracy of each model. Our findings are summarized as follows:

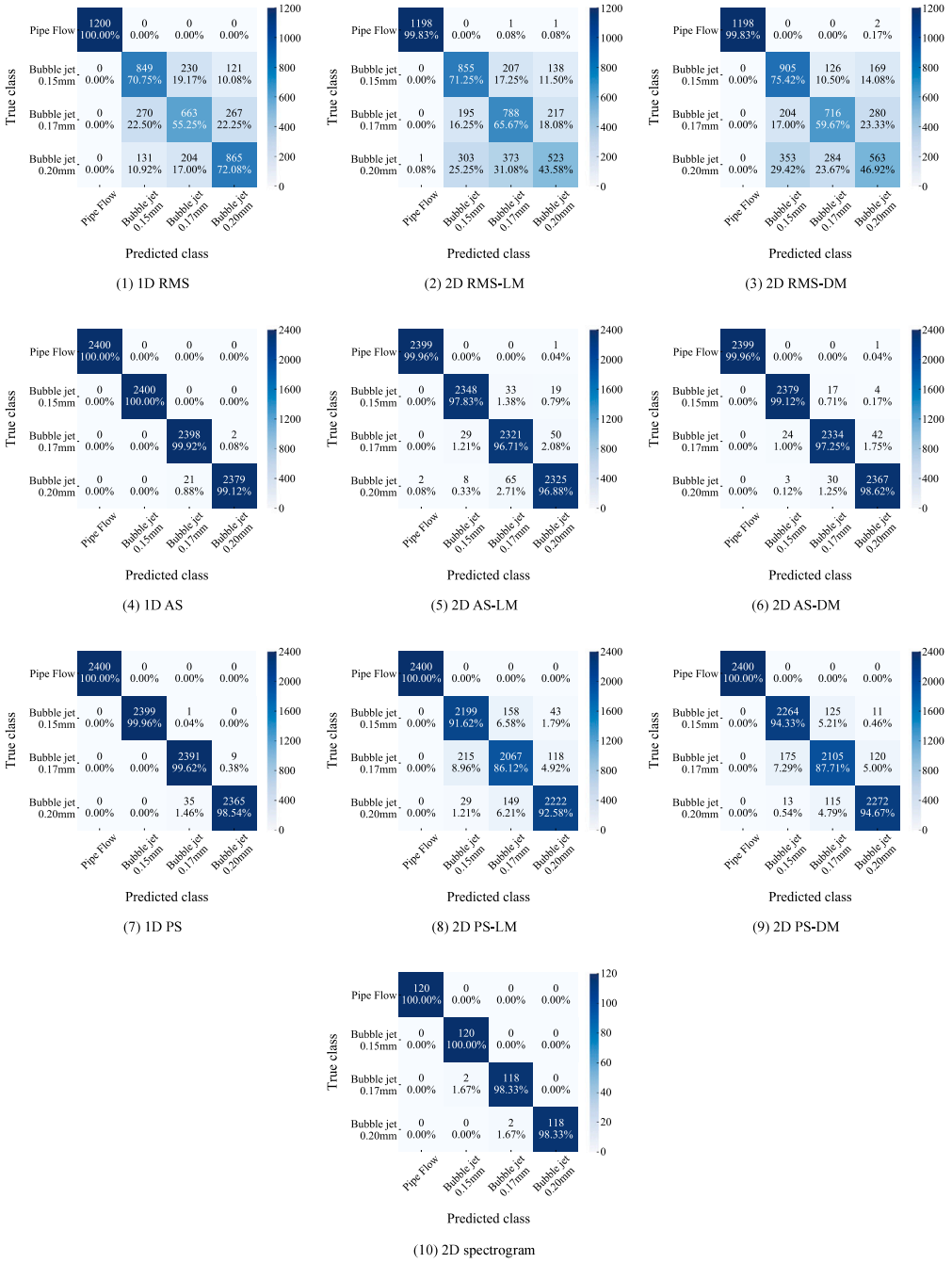


Fig. 11. Confusion matrixes of each deep learning models.

1. CNN models with the DM features we newly propose show higher performances than those with the LM features proposed in the previous study (Bae et al., 2018).
2. The deep learning models with features except (2) 2D RMS-LM, (3) 2D RMS-DM, (5) 2D AS-LM, and (6) 2D AS-DM can detect the bubble jet flow sound with an accuracy of 100.00%
3. The deep learning model with the feature of (4) 1D AS can classify the pipe flow sound, and three types of bubble jet flow sound with an accuracy of 99.76%, that is, our developed method can identify several types of bubble jet flow emitted at speed of sound, or from nozzles with slightly different diameters. This demonstrates that the method based on acoustic recognition and deep learning has great potential to sense the state of bubble jet flow in the actual SFR.

In future works, we plan to evaluate other visual representation images of acoustics, which may improve the performance of deep learning models. Also, we would like to increase the size of 2D images and employ typical convolutional neural networks such as AlexNet, VggNet, and ResNet. In addition, although the present study is positioned as a basic study, it is necessary to conduct experiments under conditions close to those in actual SFRs such as sound acquisition of the sodium–water reaction or the two-phase jet flow under higher pressure; the two-phase jet flow involves the sodium–water reaction in actual

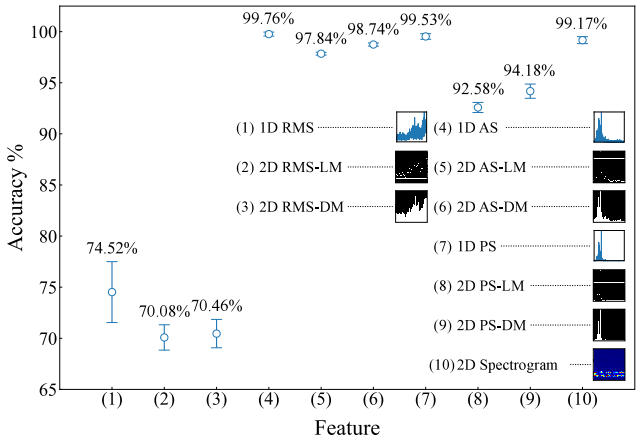


Fig. 12. Average accuracy of each deep learning model.

accidents and the pressure inside heat transfer tubes is several tens of atmospheres in SFRs.

CRediT authorship contribution statement

Nao Mikami: Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Yoshitaka Ueki:** Conceptualization, Methodology, Investigation, Resources, Writing – review & editing, Project administration, Funding acquisition. **Masahiko Shibahara:** Methodology, Resources, Supervision. **Kosuke Aizawa:** Conceptualization, Methodology, Investigation, Resources, Project administration, Funding acquisition. **Kuniaki Ara:** Conceptualization, Methodology, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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